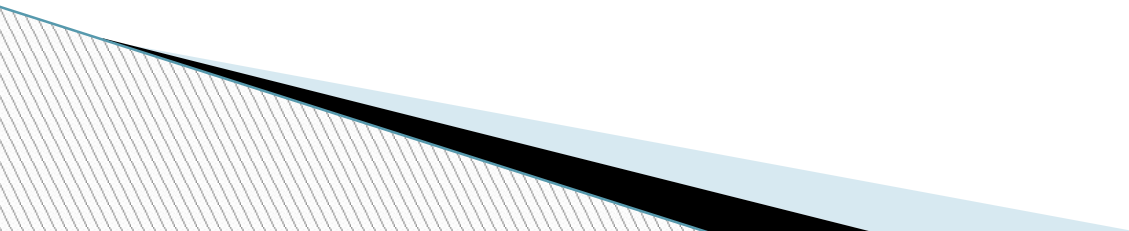


Neck trauma

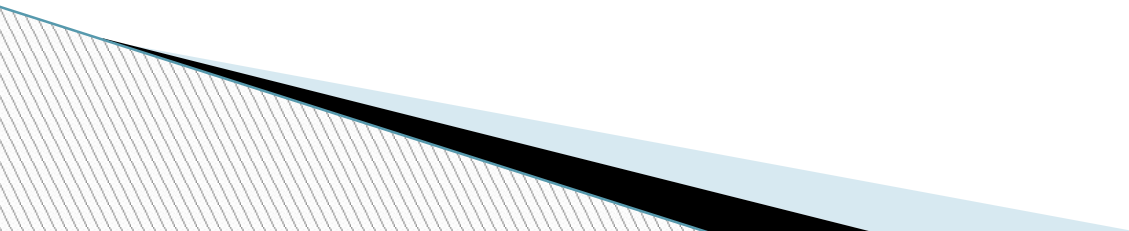
DR:Mariwan latif fatah

M.B.ch.B F.I.C.M.S



INTRODUCTION

- Neck injuries are a relatively uncommon presentation in the emergency departments, with variation across different geographical areas owing to differences in patient demographics.
- They account for 5–10% of all trauma. Their evaluation and management can be challenging, especially for the inexperienced doctor, because of the complexity of the neck anatomy, which includes vital structures in a confined space



- ▶ Introduction of diagnostic tools such as flexible endoscopy, high quality oesophagography, angiography and duplex Doppler have improved nonoperative evaluation of aerodigestive and vascular injuries.

THE TRAUMA PATIENT: GENERAL PRINCIPLES

Primary survey (survival assessment)

- ▶ The primary survey is an initial assessment of factors that cause early deaths in trauma patients.
- ▶ It focuses on factors such as **hypoxia, hypovolaemia, tension pneumothorax and head injury.**

AIRWAY AND CERVICAL SPINE PROTECTION

- ▶ The patient is examined for evidence of airway obstruction, which may present as stridor and use of accessory muscles.
- ▶ A chin lift and jaw thrust procedure should be performed.



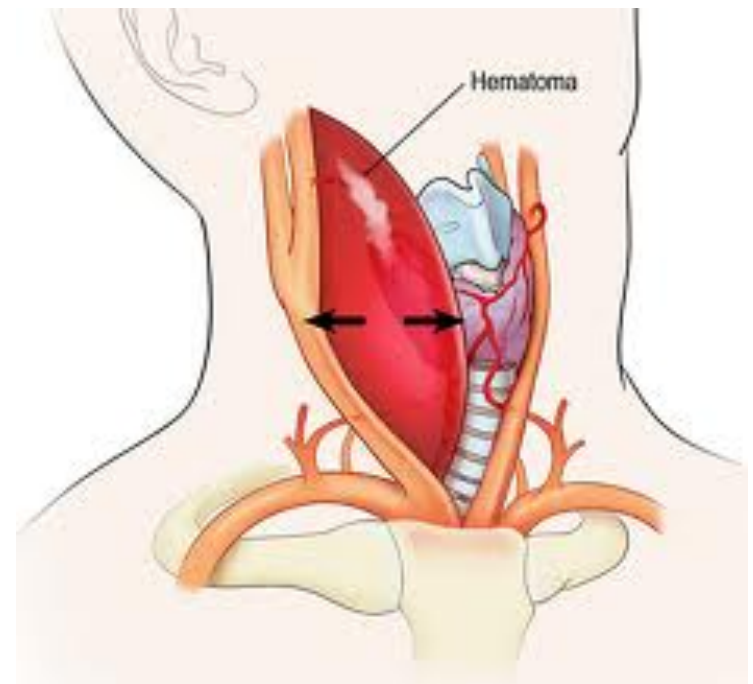
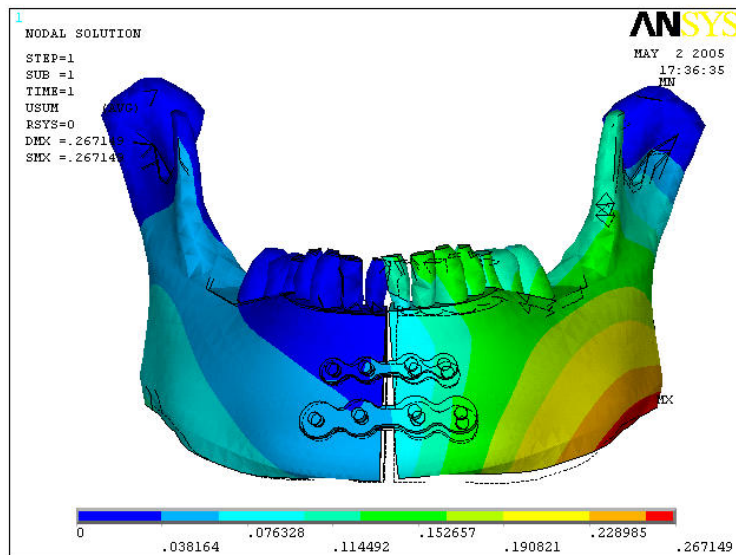
- ▶ A cervical collar or sandbags should be used to stabilize the neck until cervical spine injury is excluded.
- ▶ Patients with Glasgow coma scores (GCS) of < 8 usually require intubation.

Table 1: THE GLASGOW COMA SCALE AND SCORE

Feature	Scale Responses	Score Notation
Eye opening	Spontaneous	4
	To speech	3
	To pain	2
	None	1
Verbal response	Orientated	5
	Confused conversation	4
	Words (inappropriate)	3
	Sounds (incomprehensible)	2
	None	1
Best motor response	Obey commands	6
	Localise pain	5
	Flexion – Normal	4
	– Abnormal	3
	Extend	2
	None	1
TOTAL COMA 'SCORE'		3/15 – 15/15



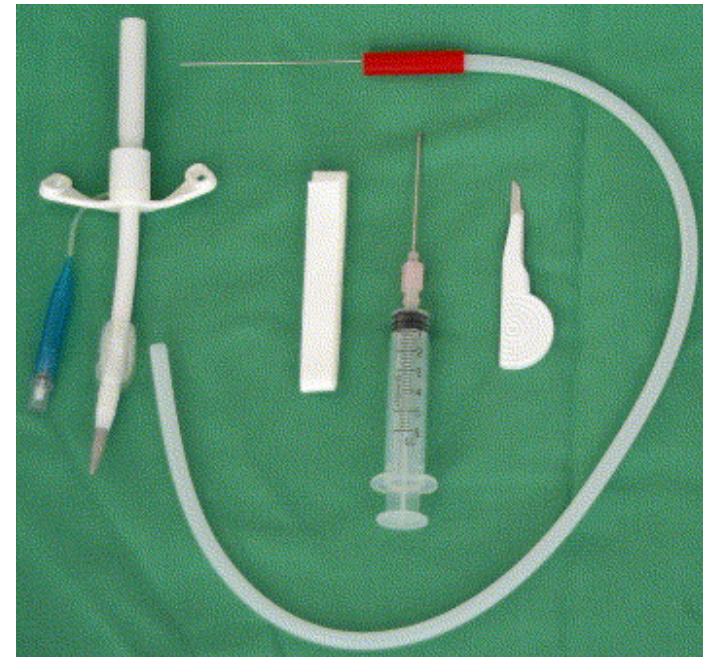
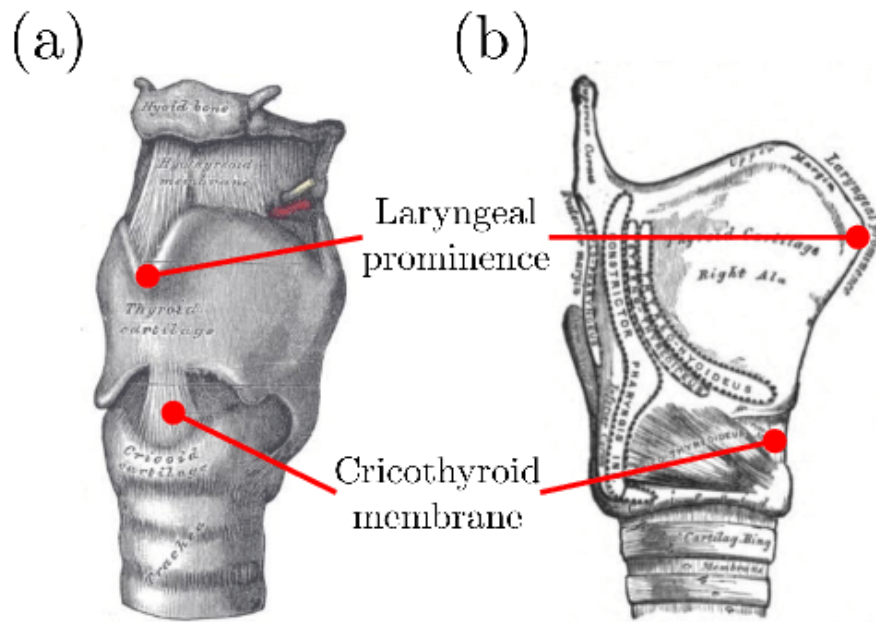
- ▶ Major facial fractures in particular mandibular fractures and large cervical haematomas may compromise the airway.



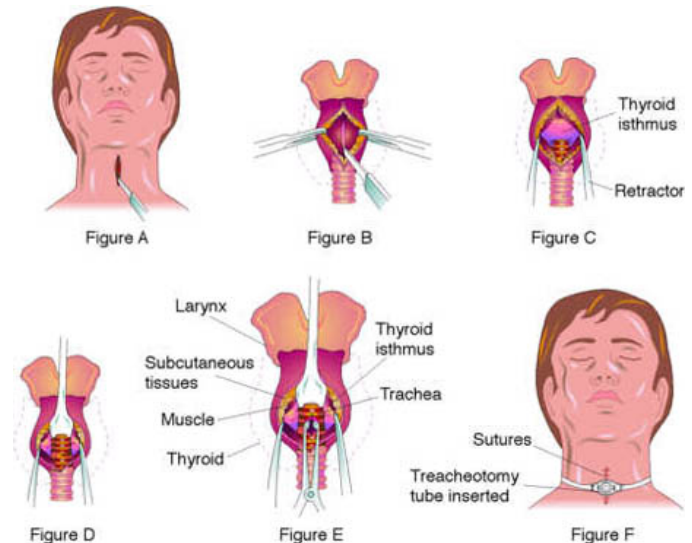
- ▶ Patients with transcervical gunshot wounds may not have airway problems initially, but must be closely observed as airway obstruction can rapidly develop.
- ▶ An endotracheal tube may sometimes be inserted directly into the trachea through the site of a penetrating cervical i



- ▶ Cricothyroidotomy must be considered if intubation fails.

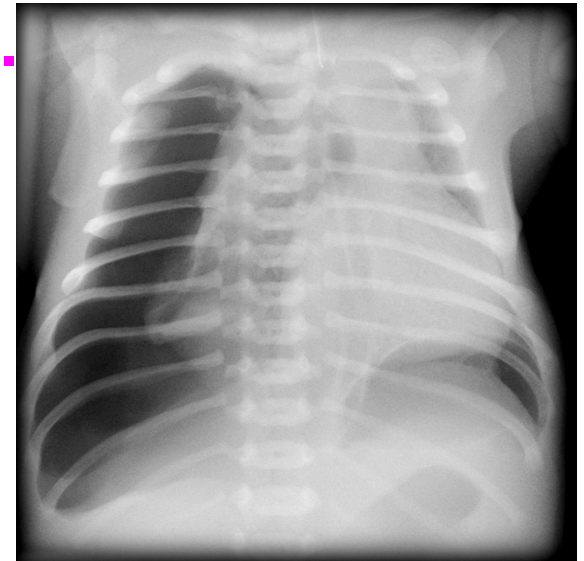


- ▶ It should be converted to a formal tracheostomy subsequently as the small tube is difficult to keep clean and in order to prevent subglottic stenosis

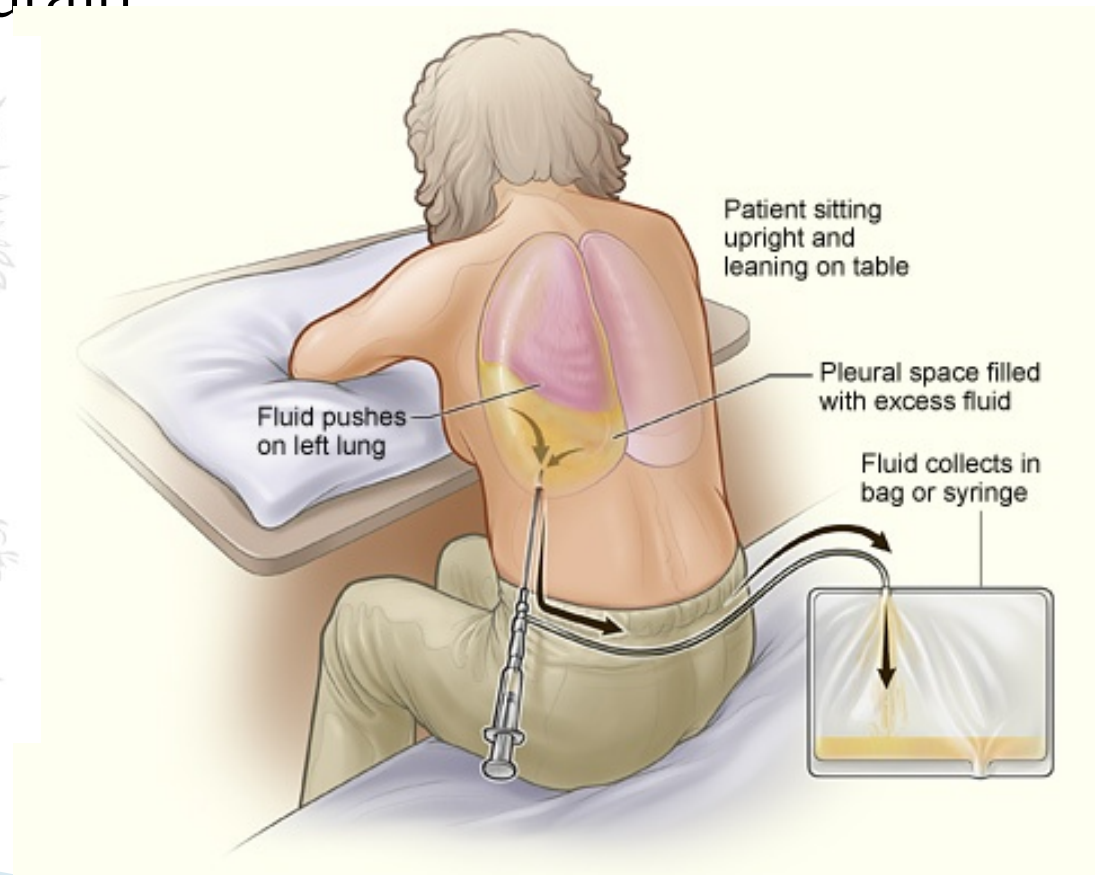
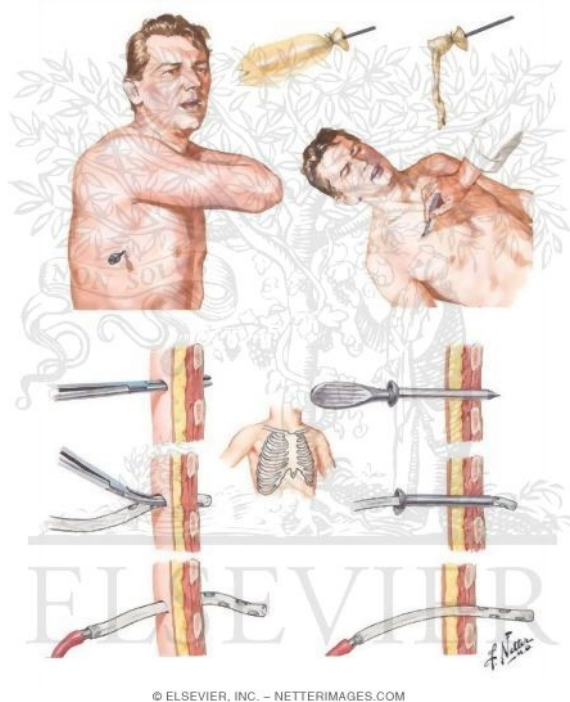


BREATHING

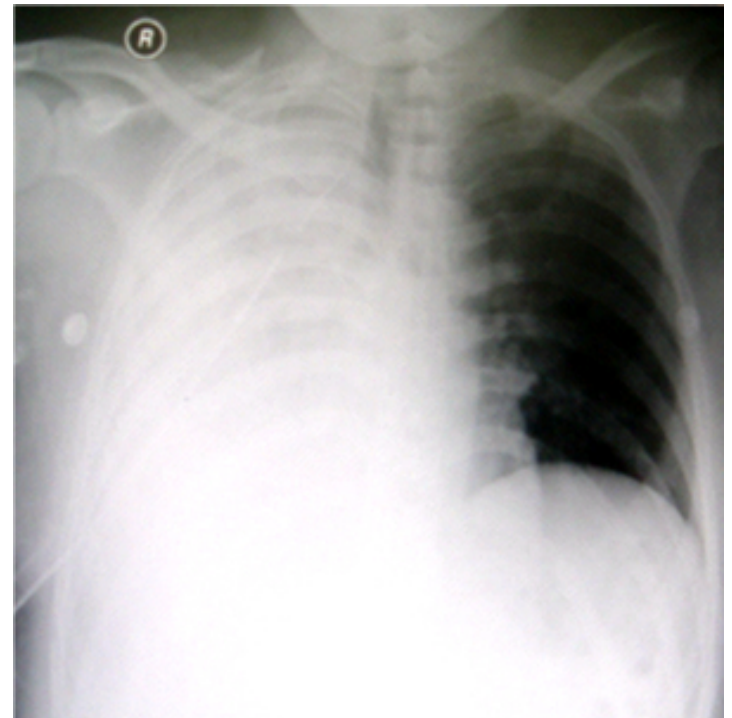
- ▶ The chest should be examined with respect to movement and air entry.
- ▶ **Tension pneumothorax presents with hypoxia, restlessness, hyper-resonance to percussion, decreased air entry, contralateral tracheal shift and elevated jugular venous pressure.**



- ▶ It is decompressed by needle thoracocentesis with a large bore needle placed through the second intercostal space in the midclavicular line, followed by insertion of a chest drain



- ▶ A large haemothorax is identified by dullness to percussion and decreased air entry, and is also managed with an intercostal drain.
- ▶ These are clinical diagnoses of life threatening conditions and immediate intervention is mandated before x-rays are obtained.



CIRCULATION AND PERFUSION

- ▶ Two high-flow lines with 14-gauge cannulae are inserted in the antecubital fossae.
- ▶ The line should not be inserted on the side of massive bleeding from a vascular injury for if there is a venous injury the fluids delivered will be bled out.



Size	Colour	Catheter (mm)		Catheter Length (mm)	Water Flow Rate (ml / min)
		I.D	O.D		
14 G	Orange	1.7	2.1	45	270
16 G	Grey	1.3	1.7	45	180
17 G	White	1.1	1.5	45	125
18 G	Green	0.9	1.3	32 / 45	80
20 G	Pink	0.8	1.1	32	54
22 G	Blue	0.6	0.9	25	33
24 G	Yellow	0.5	0.7	19	20

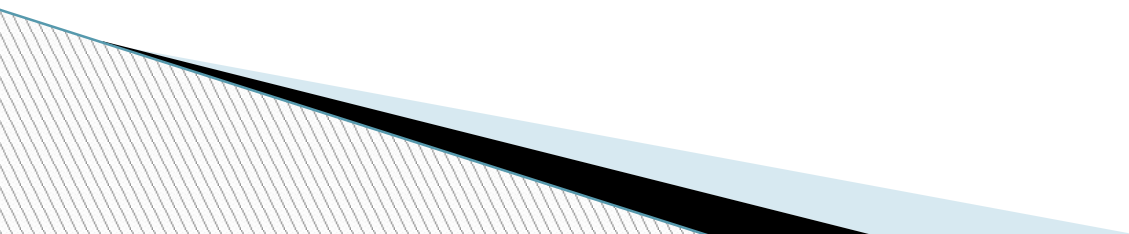
- ▶ In the shocked patient start with 2 L of crystalloid and proceed to emergency blood (O negative) if the blood pressure does not improve and cross match blood.



- ▶ A shocked patient with warm peripheries may have neurogenic shock secondary to spinal cord injury.
- ▶ Active bleeding from a cervical wound may be controlled with a compressive dressing or digital pressure.
- ▶ Failing this a large Foley catheter may be inserted into the wound the bulb inflated and the catheter cross clamped to stop blood pc



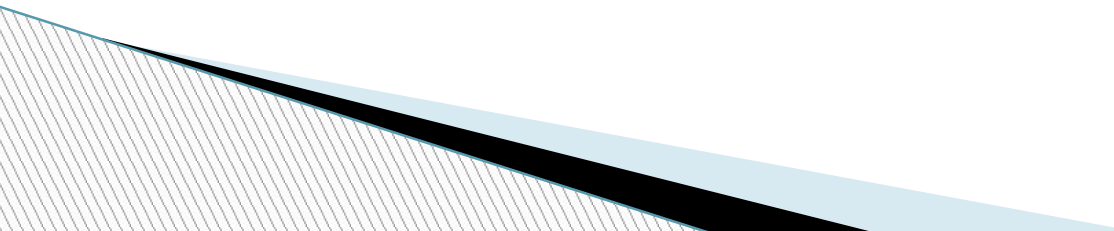
- ▶ This can be a very effective form of haemostasis, particularly with bleeding from the subclavian and cervical vessels.
- ▶ The patient should remain recumbent, or the neck wound must be covered with an occlusive dressing when the patient sits up, to prevent air embolism.



DISABILITY

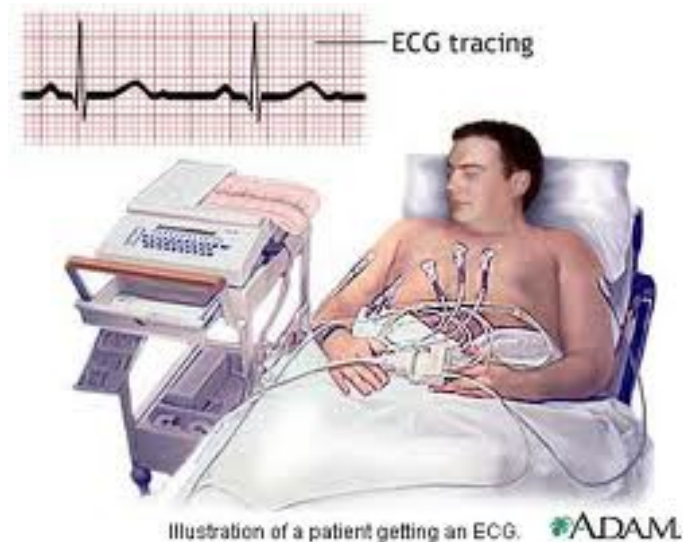
- ▶ The GCS and pupil size are recorded, and power in the limbs assessed.

EXPOSURE

- ▶ All clothes are removed in order to avoid missing associated injuries and the patient is kept warm.
- 

ADJUNCTS TO PRIMARY SURVEY

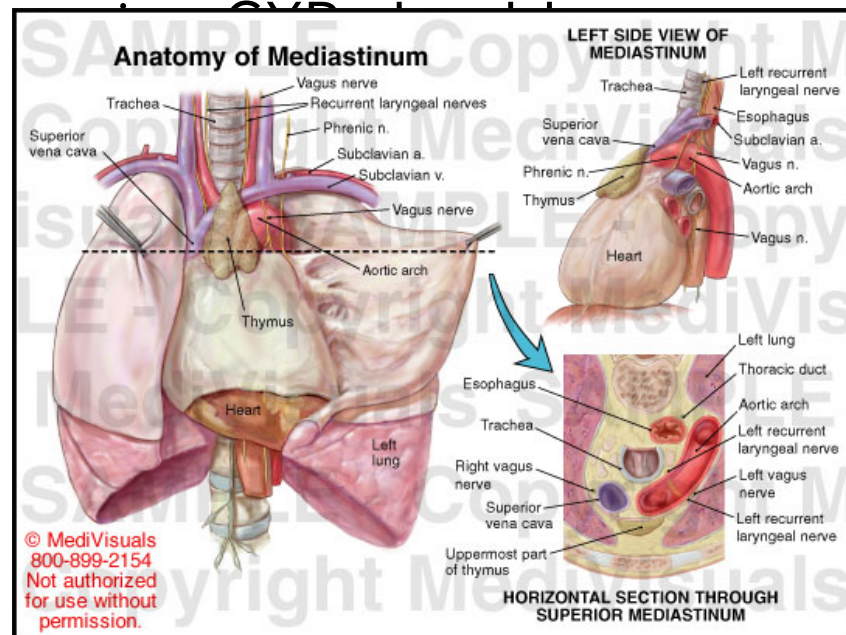
- ▶ Adequate monitoring including electrocardiogram, pulse oximetry and non-invasive blood pressure monitoring is instituted.



- ▶ The haemoglobin level, blood glucose and arterial blood gas are checked.
- ▶ Essential xrays are requested.
- ▶ In the polytrauma patient this comprises cervical spine, chest x-ray (CXR) and pelvic x-rays.
- ▶ In a patient with isolated penetrating neck trauma, a cervical spine and CXR will suffice

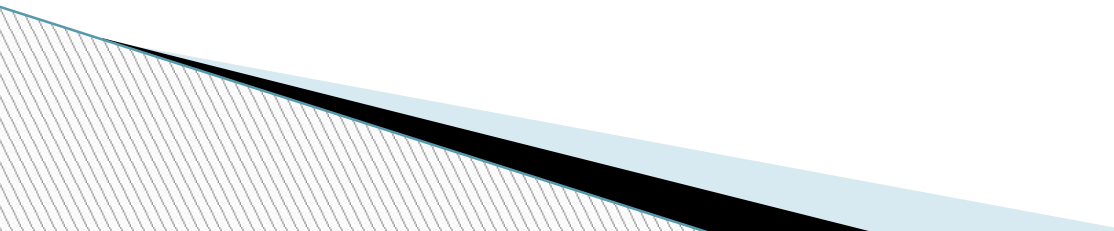


- ▶ Non urgent x-rays should be requested only on completion of the secondary survey.
- ▶ Haemo- or pneumothorax, pneumomediastinum (tracheal or oesophageal injury) and widened mediastinum should be excluded.
- ▶ The mediastinal width on a



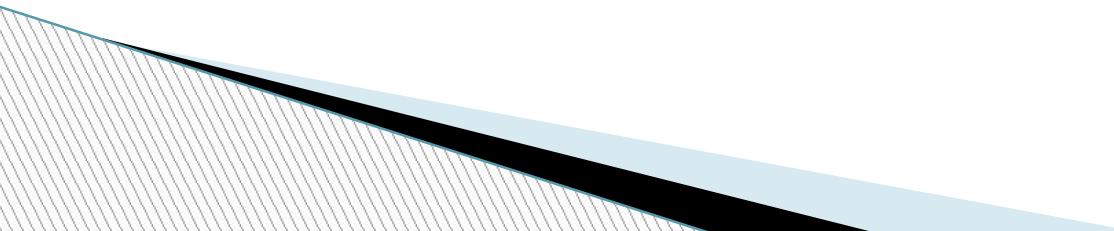
- ▶ A widened mediastinum is suggestive of a major intrathoracic injury and must be investigated further.
- ▶ Cervical spine x-ray is used to exclude spinal column injury and prevertebral air (pharyngeal or oesophageal injury) .

Secondary survey

- ▶ A detailed head-to-foot examination of the patient should be conducted.
 - ▶ In fire arm injuries the entrance and exit wounds must be noted and the tract of the projectile determined to identify which anatomical structures may have been injured.
- 

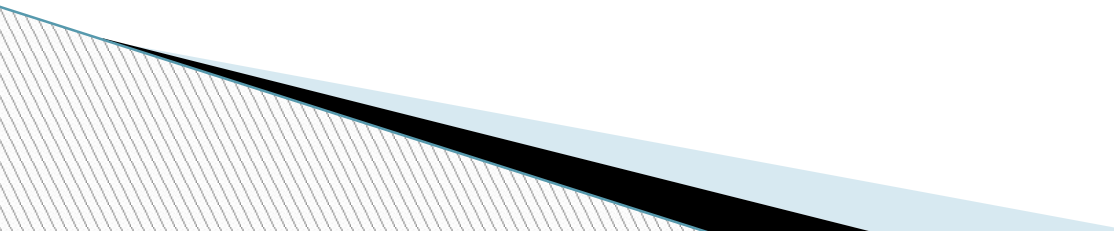
- ▶ If an exit wound is not present, then xrays are obtained (anteroposterior and lateral) to locate the bullet so that the tract of the bullet can be determined.
- ▶ The cranial nerves should be examined and Horner's syndrome excluded.

- ▶ Note the presence of large cervical haematomas, subcutaneous emphysema, the jugular venous pressure, the presence of blood in the nasogastric tube and tenderness over the mandible.

- ▶ Wounds should **not** be probed as this may result in massive bleeding from an arterial injury.
 - ▶ The peripheral pulses should be checked for discrepancy or absence.
- 

- ▶ The abdomen and pelvis should be examined, followed by a full neurological examination during which spinal cord and brachial plexus trauma are excluded.
- ▶ Note the presence of hemiplegia (hemitransection of spinal cord).
- ▶ The patient should be log-rolled in order that the back can be examined for trauma.

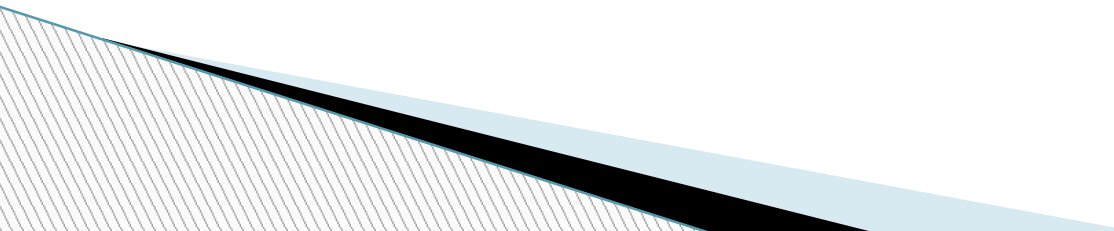
HISTORY

- ▶ Take a history with specific reference to the nature of the trauma.
 - ▶ Enquire about symptoms related to oesophageal injury (**haemoptysis, haematemesis, dysphagia and odynophagia**) and recurrent laryngeal nerve or laryngeal injury (**dysphonia, stridor**),
 - ▶ Past medical and surgical history are documented, as well as the time of the patient's last meal.
- 

FURTHER INVESTIGATIONS

- ▶ Any further x-rays that are required after the secondary survey are now completed. Special investigations such as angiography, computed tomography (CT) scan, ultrasound and gastrograffin/barium studies may now be performed in a haemodynamically stable patient.

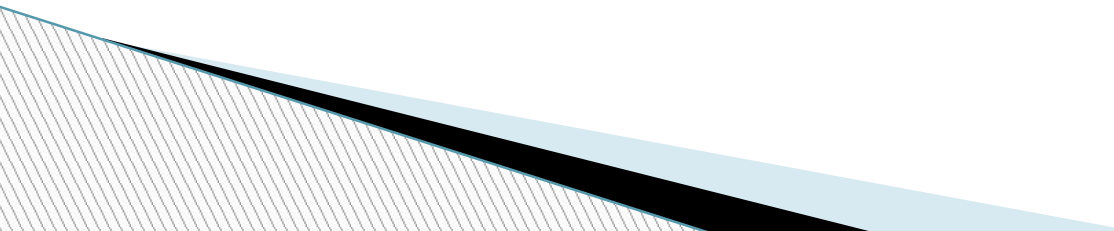
FULL DOCUMENTATION

- ▶ All injuries are documented and the patient is referred for definitive care.
- 

AETIOLOGICAL CLASSIFICATION

- ▶ Unlike stab wounds and low velocity gun shot wounds , close-range shotgun, rifle and bomb injuries cause extensive soft tissue trauma:
 - ▶ penetrating
 1. **stabs**
 2. **gun shot wound (GSW)**
 - low velocity
 - high velocity
 - shotgun
 - ▶ blast
 - ▶ blunt

ZONES OF THE NECK

- ▶ Roon and Christensen's classification is most commonly used for description of the site of cervical trauma .
 - ▶ The neck contains three principal anatomic components:
 1. **vascular (carotid, vertebral and subclavian arteries, jugular and subclavian veins)**
 2. **digestive (pharynx and oesophagus)**
 3. **respiratory (larynx and trachea).**
- 

- ▶ All three zones contain major vascular and aerodigestive structures.
- ▶ However injuries to zones I and III are both diagnostically and surgically challenging.

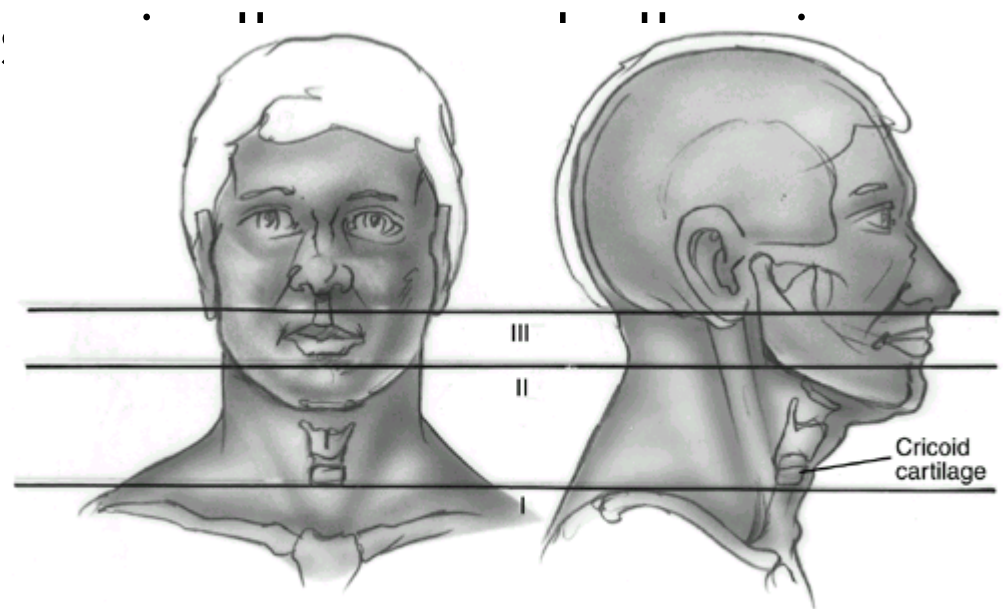
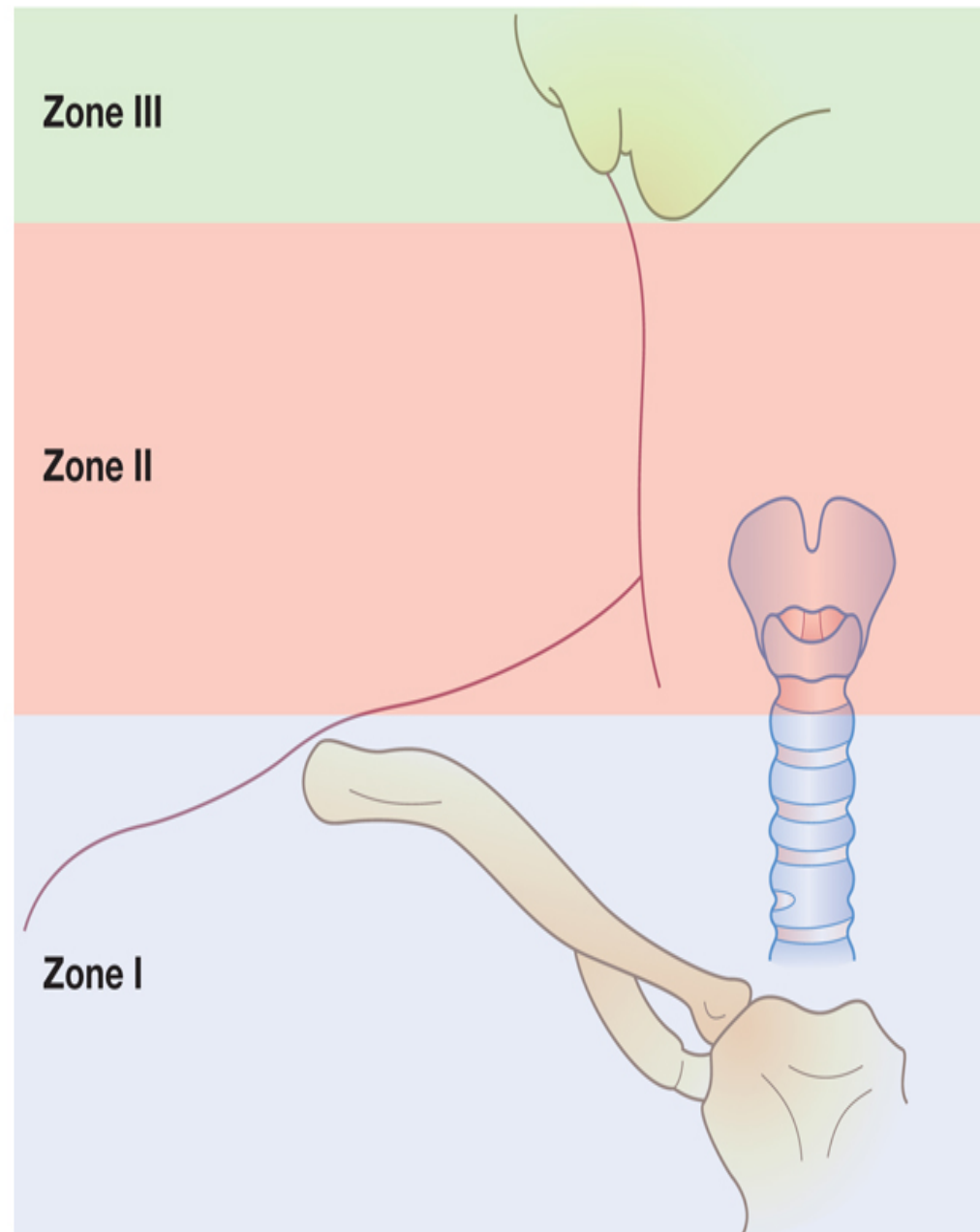
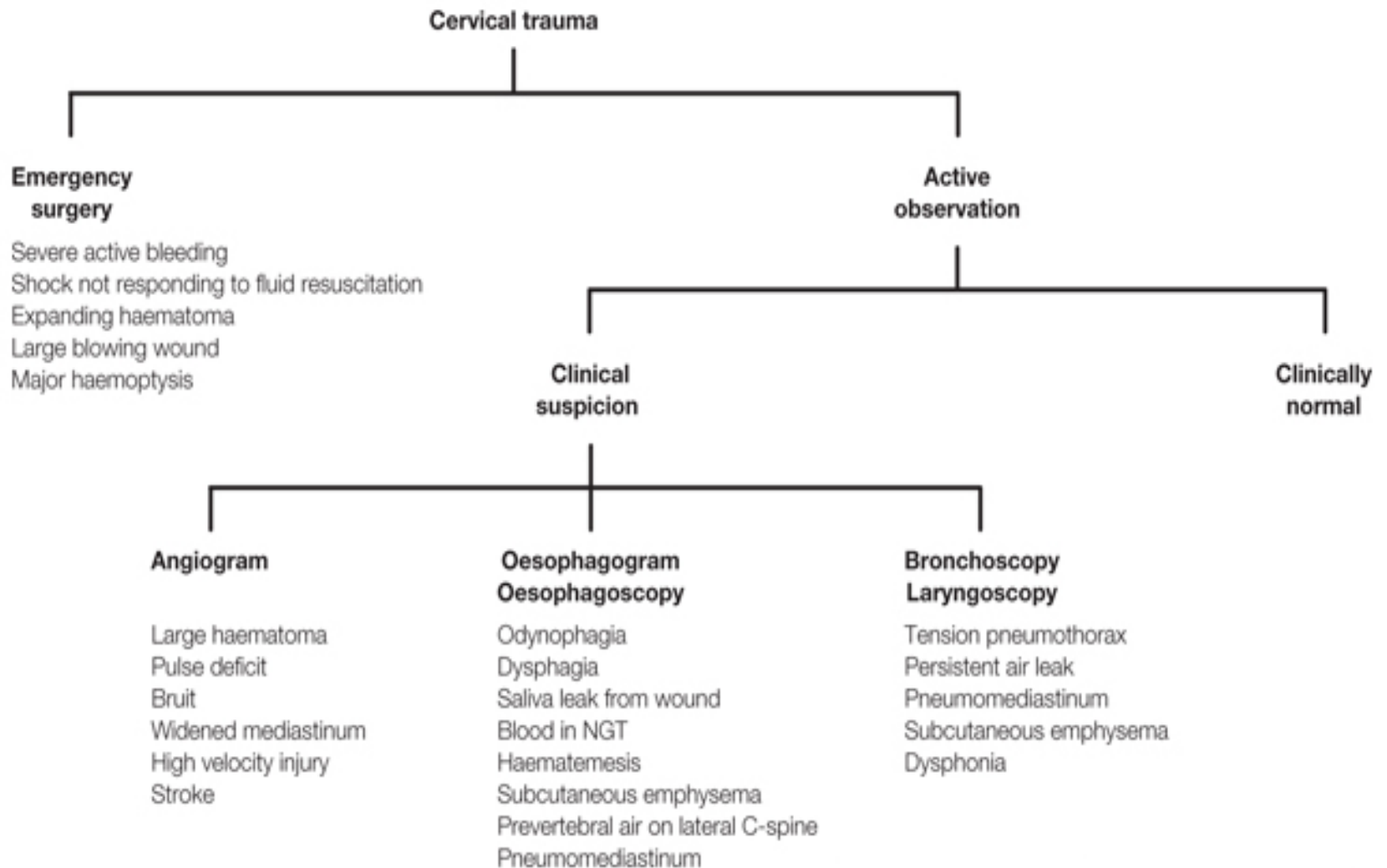


FIG. 73-8. Horizontal entry zones of the neck for penetrating injuries to the neck. (Modified from Jurkovich GJ. The neck. In: Early care of the injured patient. Moore EE, ed. Toronto: BC Becker, 1990:126.)

- Roon and Christensen classification

1. **Zone I**: sternal notch/clavicle to cricoid cartilage.
2. **zone II**: cricoid cartilage to angle of mandible.
3. **zone III**: angle of mandible to base of skull.

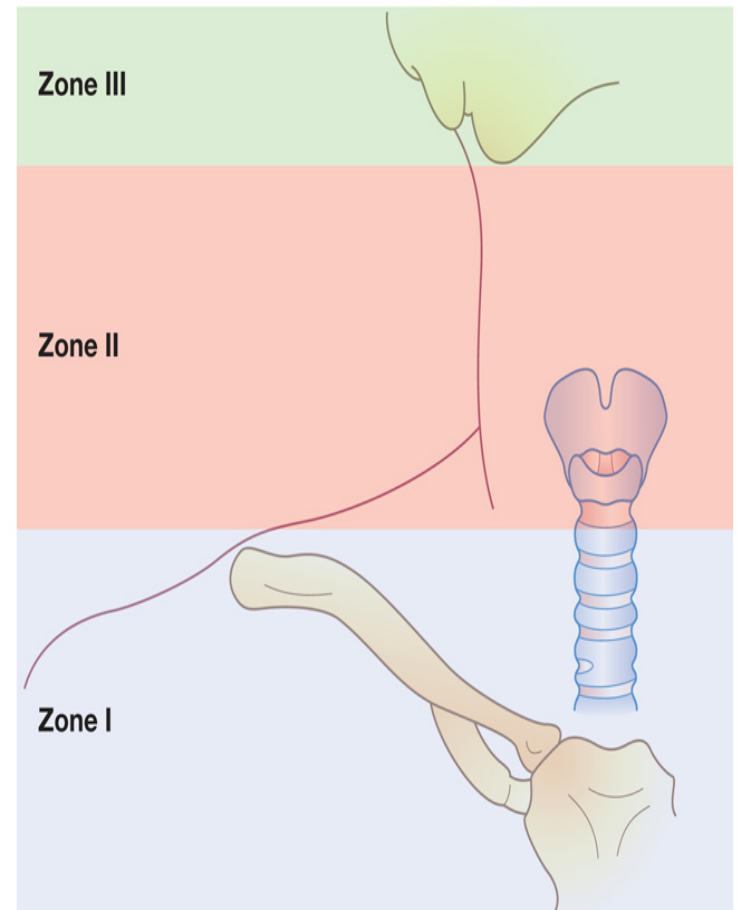




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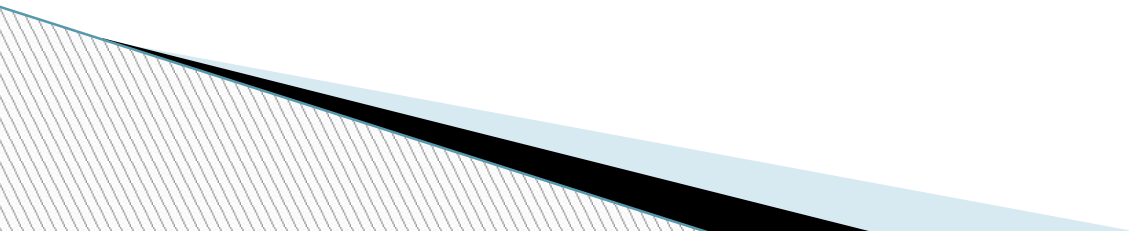
(Selective surgical exploration algorithm)

- ▶ Vascular injuries in zones I and III can be difficult to assess clinically.



Reference

- ▶ Scott brown-ENT textbook
- ▶ Logan turners – disease of ear , nose and throat



THANKS